Stochastic Processes 2021/22 Semester 1

Tutorial 6 Extension Questions

Please note: These questions are an extension to Tutorial 6 for individual training only and are not supposed to be discussed during tutorials or office hours. Answers are available for all questions, but worked solutions are only available for some hard questions.

1. Consider the three i.i.d.r.v.'s, X_1, X_2, X_3 provided in Tutorial 6 Question 4. Find:

   (a) $\mathbf{P}(X_1 - X_2 - X_3 = 0)$.

   (b) $\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = 2k)$.

   (c) $\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = 2l)$.

   (d) Whether the value of sum X_1, X_2, X_3 depends on p? Whether the probability $\mathbf{P}(X_1 = k \mid X_1 + X_2 + X_3 = n)$ $(n > k)$ depends on p?

2. Y_1 and Y_2 are following the Exponential distributions shown in Tutorial 6 Question 5. Find distribution functions of r.v.'s $Z = \min(Y_1, Y_2)$ and $T = \max(Y_1, Y_2) + \min(Y_1, Y_2)$.